

Verified first-shock candidate: review packet

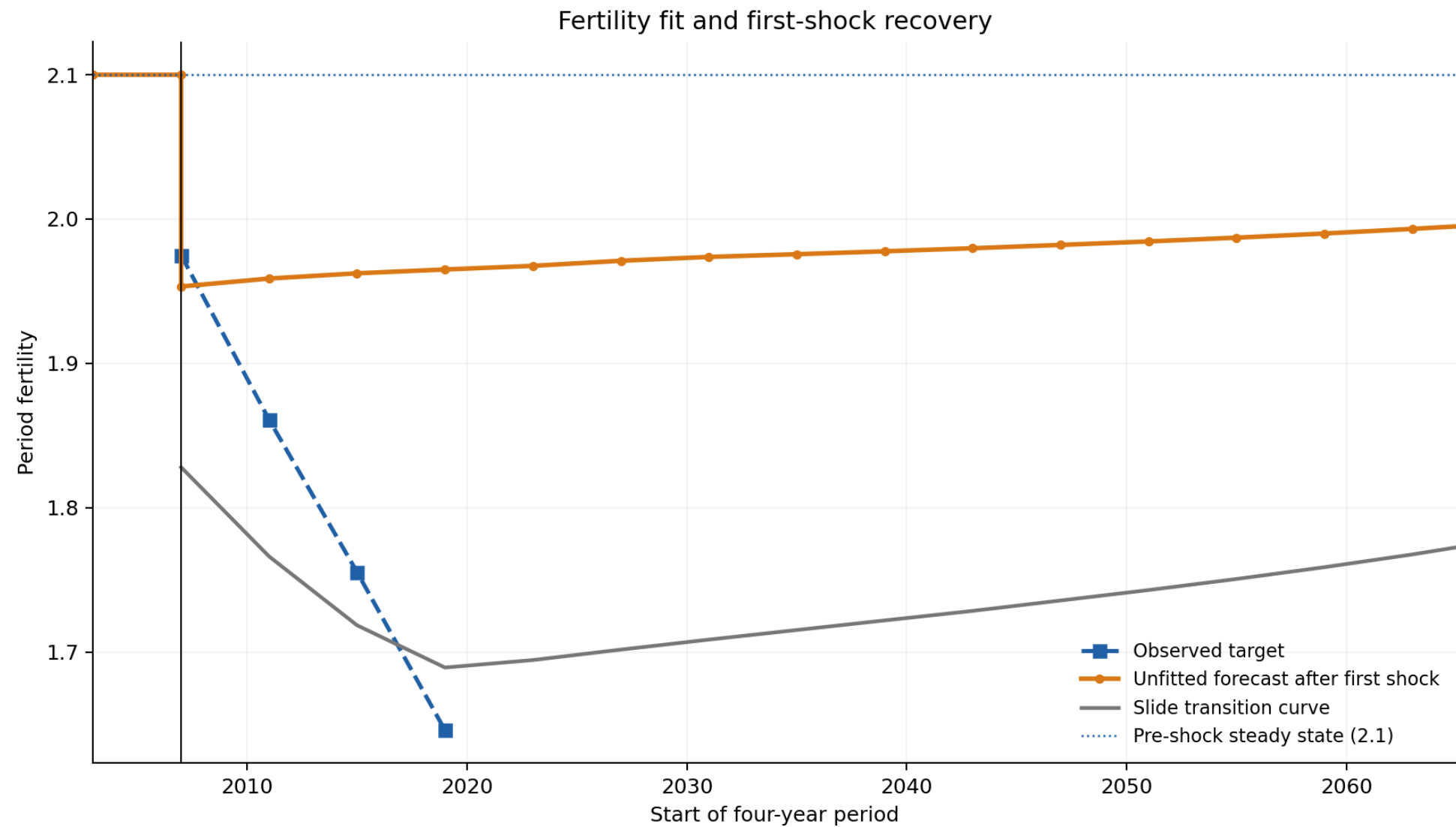
Saved results only. The 104-period forecast passes finite-horizon market and fiscal checks and reproduces exactly. It is not a completed historical fit: no shock meets both equilibrium and fertility-fit acceptance.

Preferences fall from 0.1489153 to 0.1289153 in 2007 and are expected to remain there forever. The remaining historical shocks have not been estimated. Later orange observations are forecasts conditional on that single shock.

Observed years	Data	Candidate forecast	Gap	Fit status
2008–2011	1.9749	1.9532	-0.021629	Candidate verified, fit missed
2012–2015	1.861	1.9588	0.097753	Not estimated
2016–2019	1.7554	1.9624	0.207	Not estimated
2020–2023	1.6458	1.965	0.31928	Not estimated

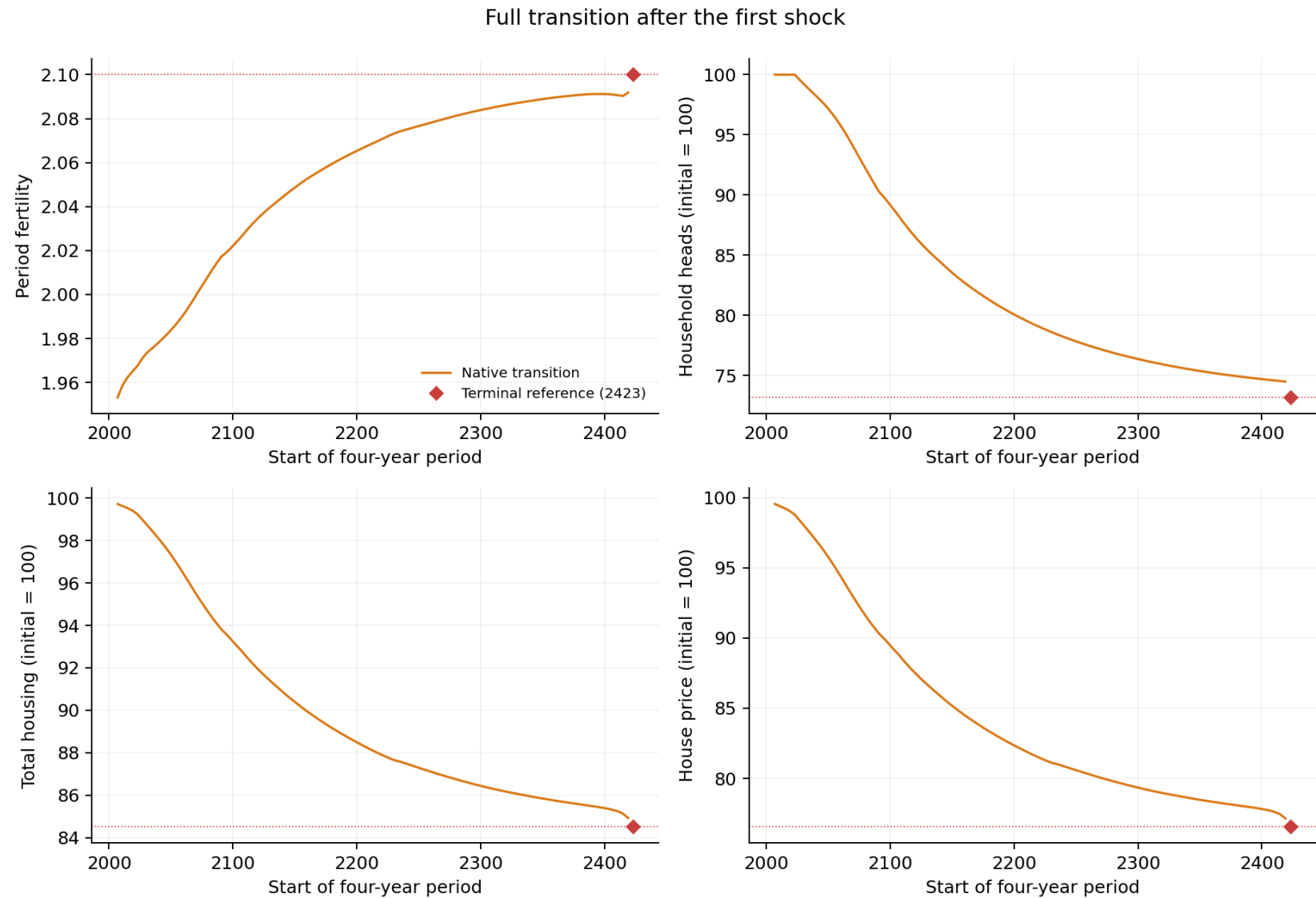
The grey comparator is the curve in the saved presentation PDF: four announced preference levels, still unconverged. Its old permanent-shock caption was inconsistent with the plotted data. The two experiments therefore differ in both shocks and expectations.

Fertility: data, slide curve and recovered candidate



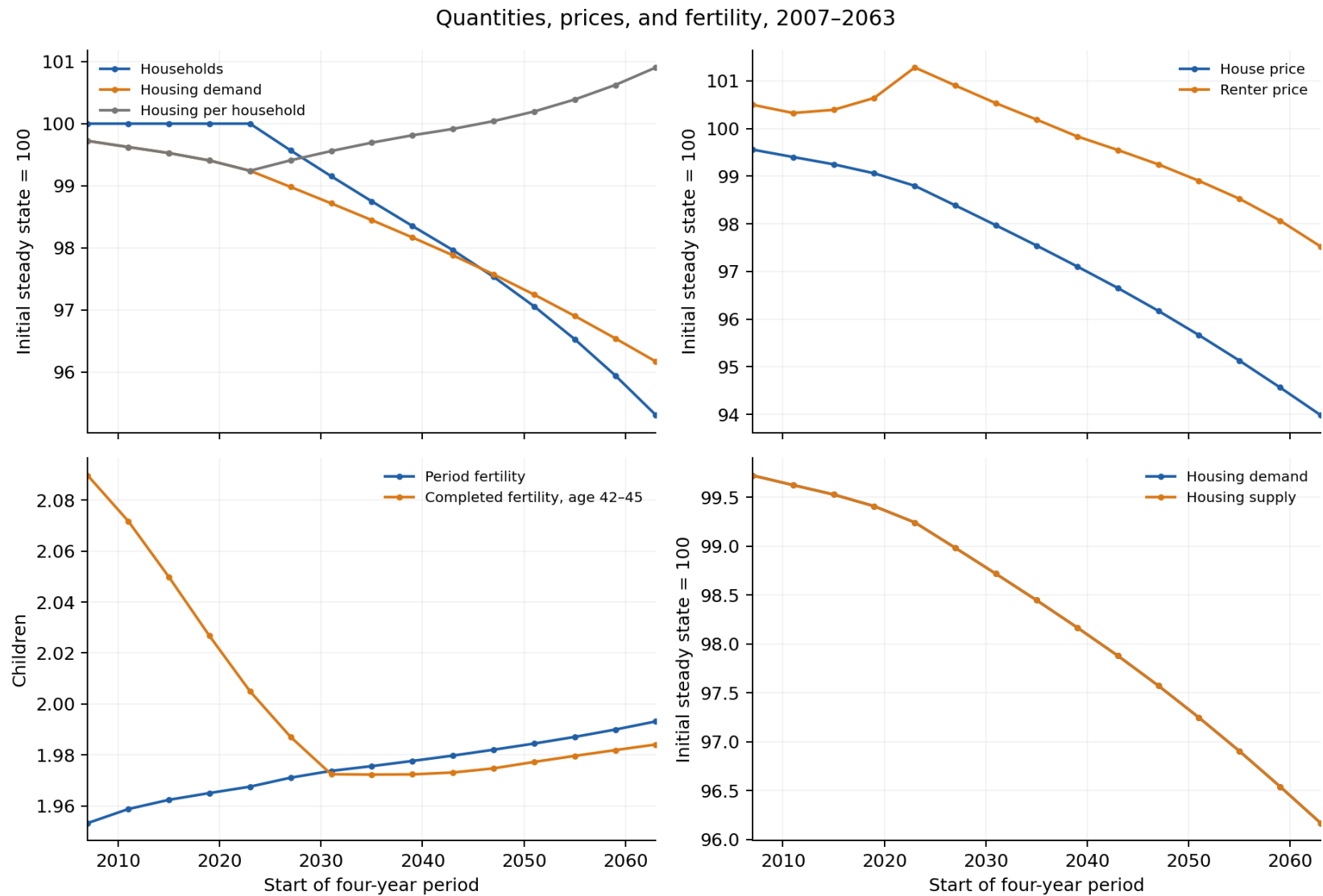
Candidate forecast after one shock; later shocks remain unfitted. Four-year model periods. Source: verified native rows and fertility observations.

Full transition and verified stationary endpoint



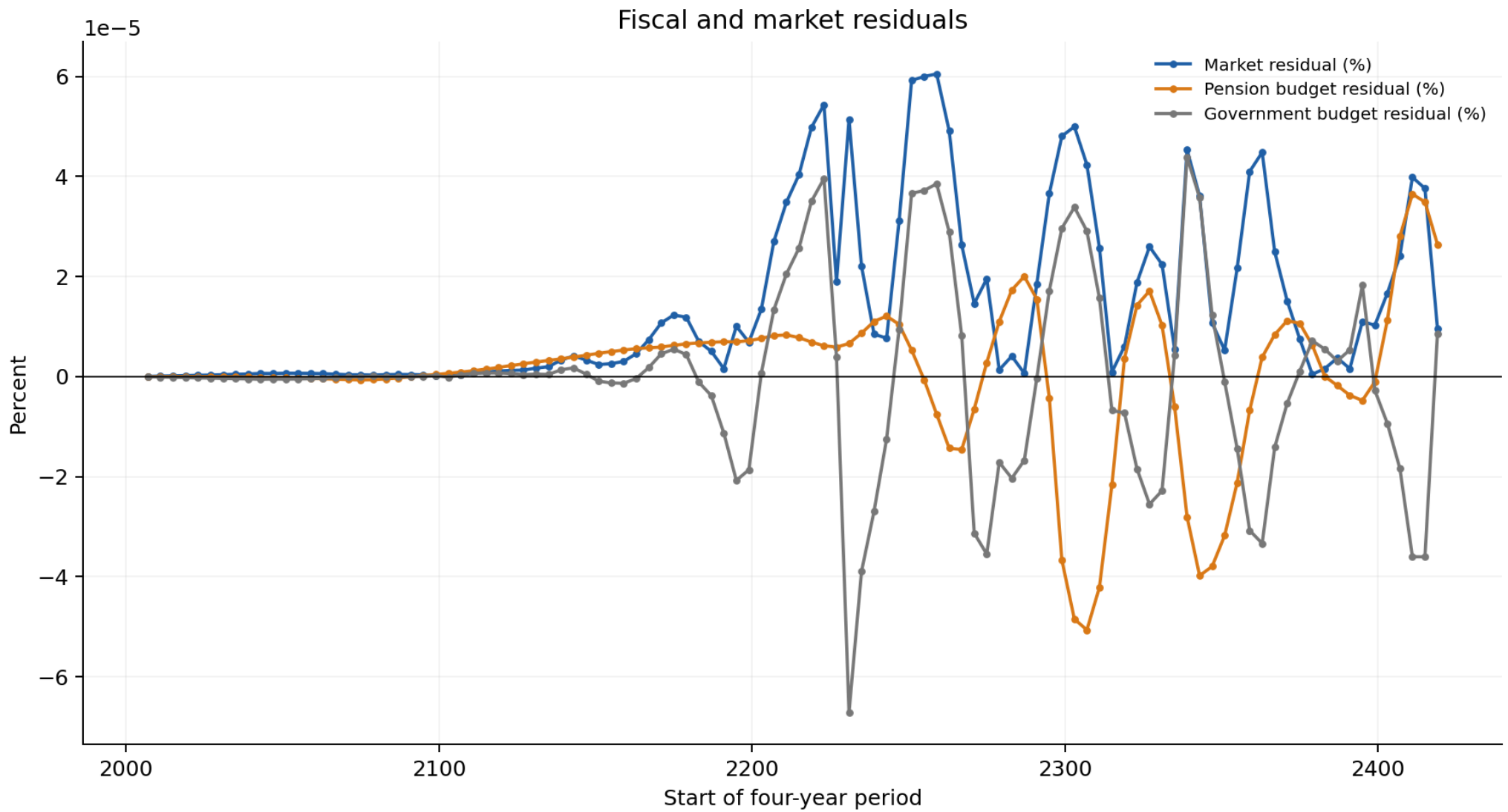
Candidate forecast after one shock; later shocks remain unfitted. Four-year model periods. Source: verified native rows and fertility observations.

Quantities, prices and reconstructed completed fertility



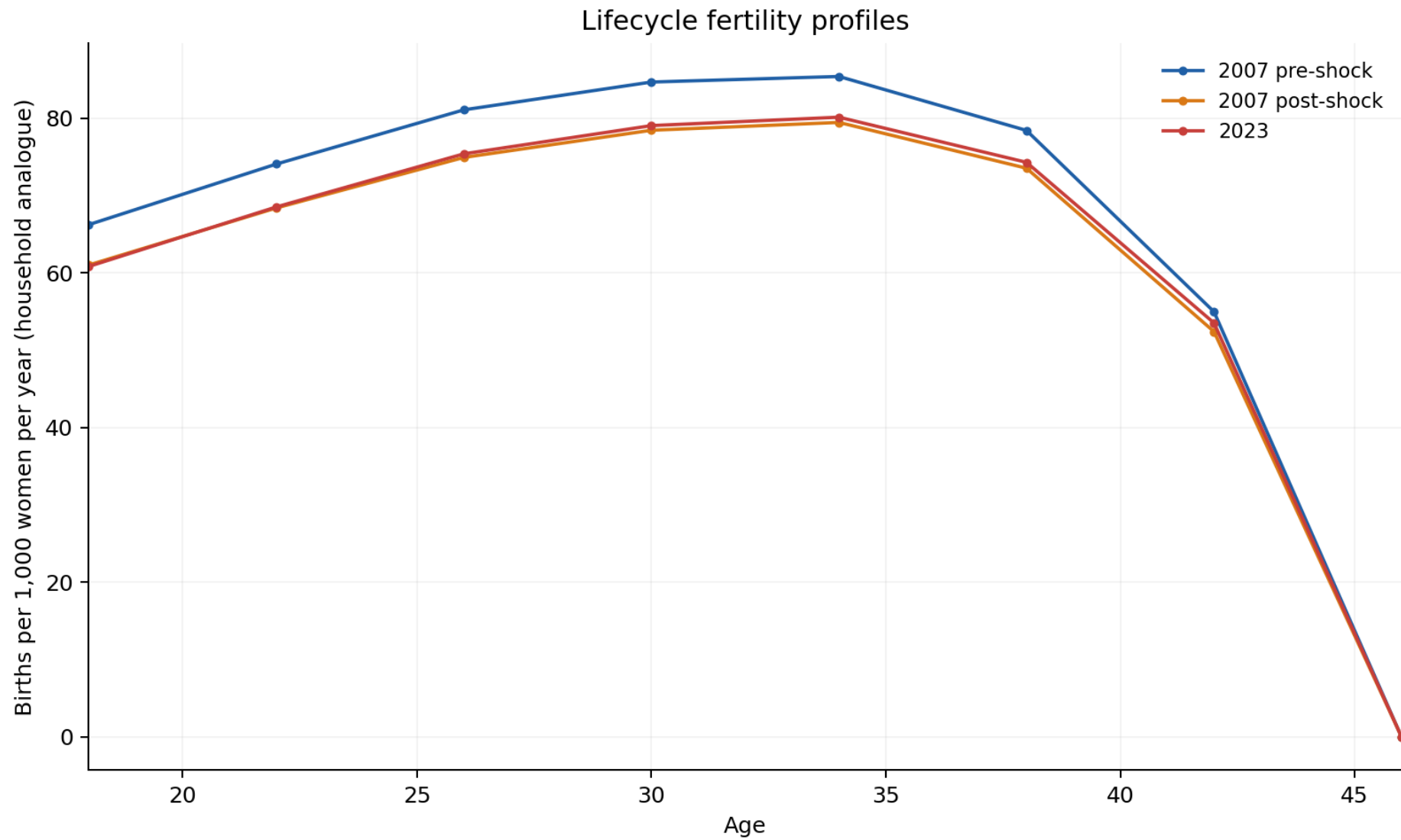
Candidate forecast after one shock; later shocks remain unfitted. Four-year model periods. Source: verified native rows and fertility observations.

Market and fiscal residuals



Candidate forecast after one shock; later shocks remain unfitted. Four-year model periods. Source: verified native rows and fertility observations.

Fertility over the lifecycle



Candidate forecast after one shock; later shocks remain unfitted. Four-year model periods. Source: verified native rows and fertility observations.

Initial parameter vector: complete empirical review

All 13 rows are retained. These describe the unchanged initial state, not a new calibration. The current provenance review labels the scored restrictions proposed rather than certified SMM estimates. The 2.1 normalization is separate and unscored; -- denotes an unavailable/not-applicable weight or contribution.

Moment	Target	Model	Gap	Weight	Contribution
Initial model completed fertility	2.1	2.1	8.2255e-07	--	--
Childless women, ages 40–44	0.19828	0.19626	-0.0020204	35532	0.14504
Exactly one child among mothers, ages 40–44	0.21366	0.21644	0.0027836	26953	0.20885
Period mean first-birth age	25.976	26.095	0.11898	139.83	1.9795
First births at age 30+	0.24928	0.23216	-0.017123	13866	4.0654
Wealth / annual gross labor earnings	6.1459	4.8934	-1.2524	7.5951	11.914
Annual bequests / aggregate wealth	0.0088	0.0085824	-0.00021759	5.1653e+06	0.24455
Old wealth/income p90 / median, ages 76–84	3.5159	4.4998	0.98387	10.616	10.277
Mean occupied rooms, capped at 9	5.5611	6.4247	0.86356	128.02	95.469
Ownership, heads 30–55	0.64833	0.5399	-0.10844	2339.4	27.508
First-birth room response, -1 to +3	0.72025	0.98081	0.26056	137.57	9.3395
Rooms: 3+ versus 1–2 resident children (model dependent proxy)	0.34707	0.17204	-0.17502	280.53	8.5935
Recent-parent ownership gap	0.1629	0.1441	-0.018793	27056	9.5553

Parameters and restrictions

Parameter	Value	Lower	Upper	Near bound	Role / restriction
beta_annual	0.99	0.94	0.99	True	structural coordinate
kappa_fert	0.33773	0.02	50	True	structural coordinate
kappa_fert_continuation	0.39786	0.02	50	True	structural coordinate
chi	1.0497	0.1	5	False	structural coordinate
H0	8.1121	0.2	80	False	structural coordinate
theta0	0.081051	0	8	False	structural coordinate
theta1	0.085204	0.02	16	True	structural coordinate
first_birth_fixed_cost	0.26576	0	8	False	structural coordinate
h_P	2.3	0.1	2.3	True	mapped first-child requirement

Annual beta is estimated subject to a 0.99 upper bound. Full source status, raw bounds and provenance are preserved in parameters.csv; this packet does not promote these values as new estimates.

Parameters and restrictions (continued)

Parameter	Value	Lower	Upper	Near bound	Role / restriction
hbar_child_rooms	0	--	--	False	
psi_child	0.14892	--	--	False	
payroll_tax	0.179	--	--	False	
pension_period	2.0464	--	--	False	
housing_supply_elasticity	0.63	--	--	False	
tenure_choice_kappa	0.005	--	--	False	
alpha_cons	0.733	--	--	False	
sigma	2	--	--	False	

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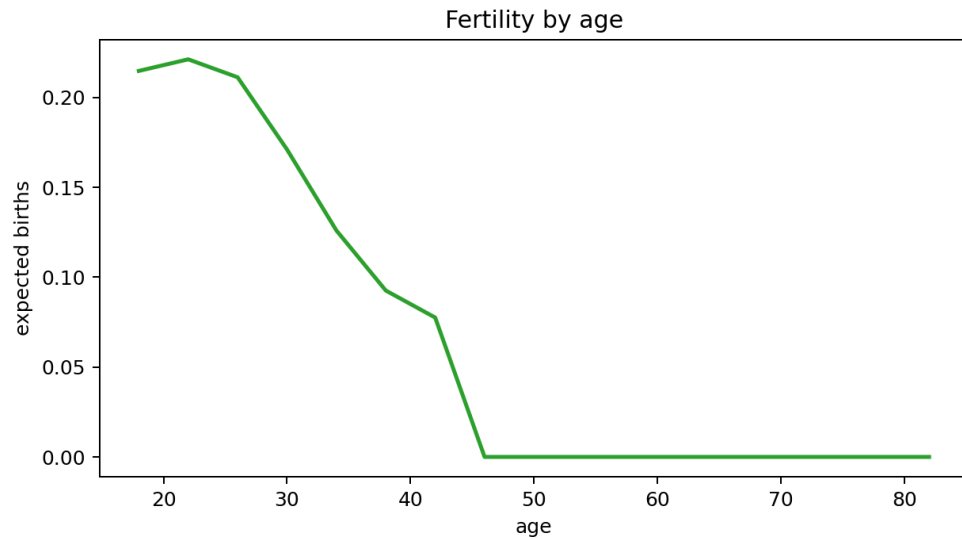
Validation status and saved-data coverage

Check / object	Result
Finite market/fiscal equilibrium	Passed for this first candidate; exact final reproduction.
Historical shock fit	Zero of four accepted. First gap exceeds 0.005 tolerance.
Verified terminal stationary equilibrium	Passed. This does not certify the path has reached it.
Terminal household mass gap	1.742%
Terminal distribution relative L1 distance	9.446%
Horizon robustness	Not established.
2023 fertility by age	Available from the candidate forecast.
2023 housing, ownership and wealth by age; intergenerational allocation	Not saved for this candidate; would require additional household-state extraction/replay.
New policy comparison	Not run: historical fit did not reach 2023.

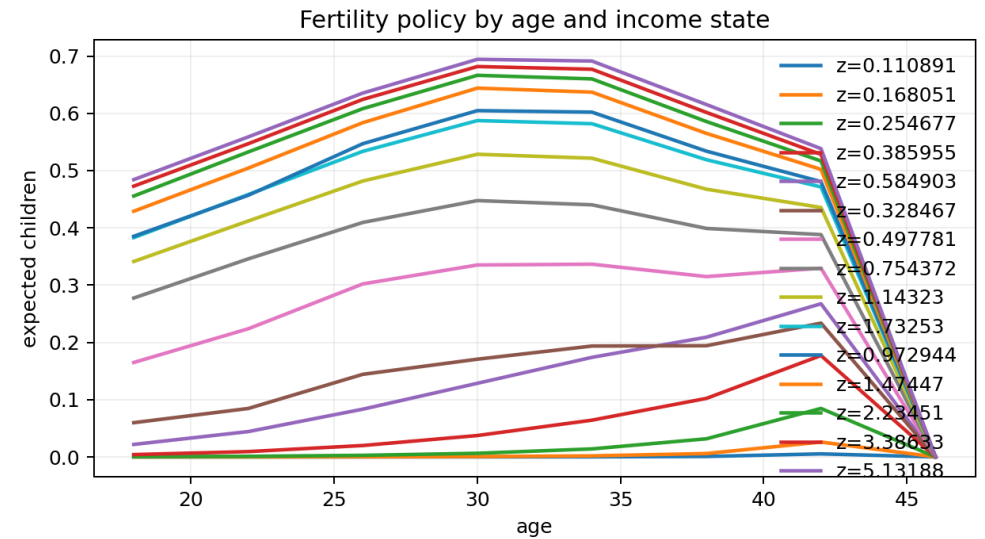
Household diagnostics: 2007 after the first shock

Saved standard plots. The legacy file lifecycle_2023.csv belongs to this 2007 state. Expected-choice statistics in the legacy fertility panels are distinct from the measured all-birth age rates above; summary.json's tfr field measures completed fertility.

fertility by age



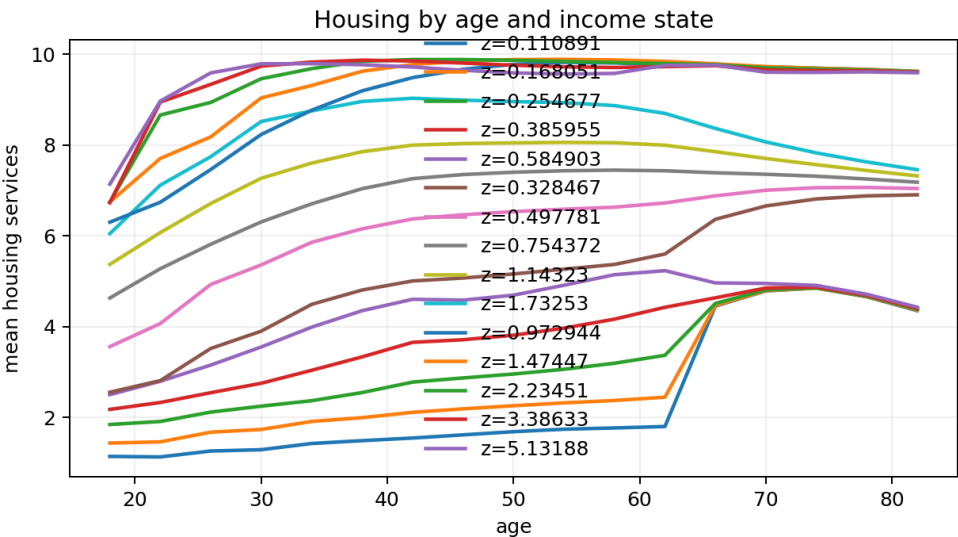
fertility policy by age income state



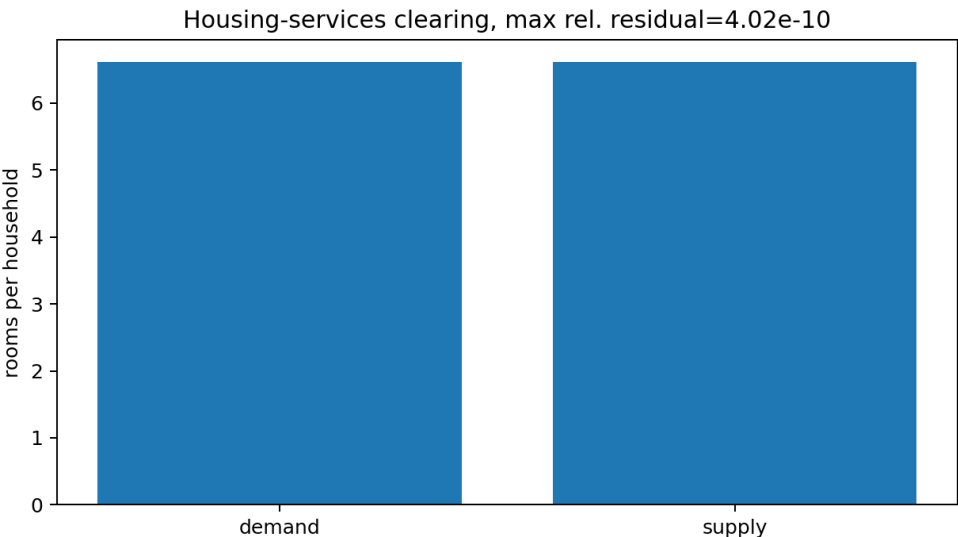
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housing by age income state



housing market



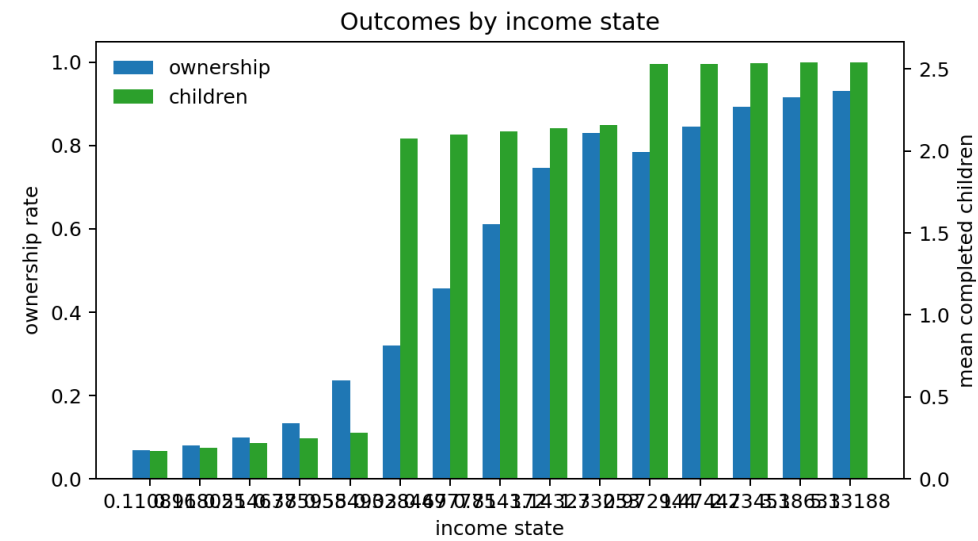
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housing prices



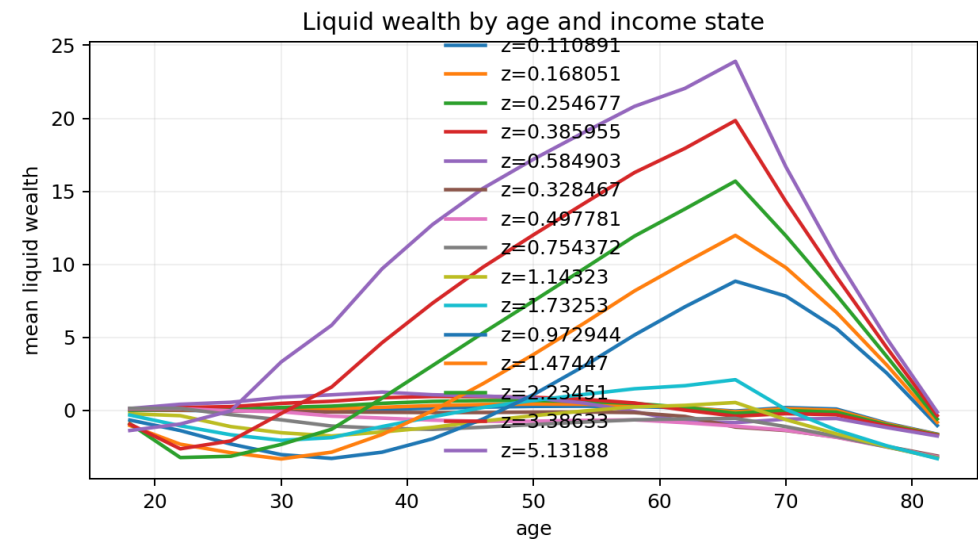
income state outcomes



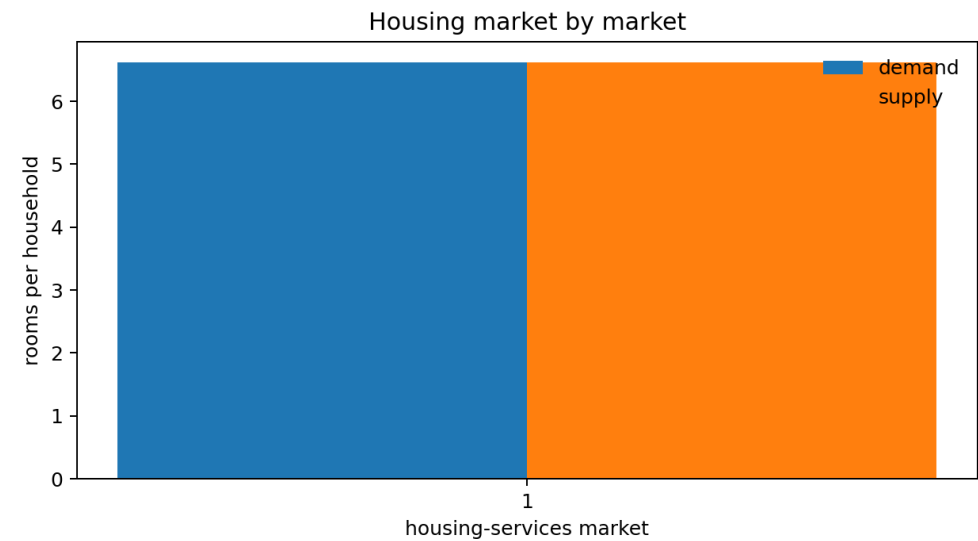
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liquid wealth by age income state



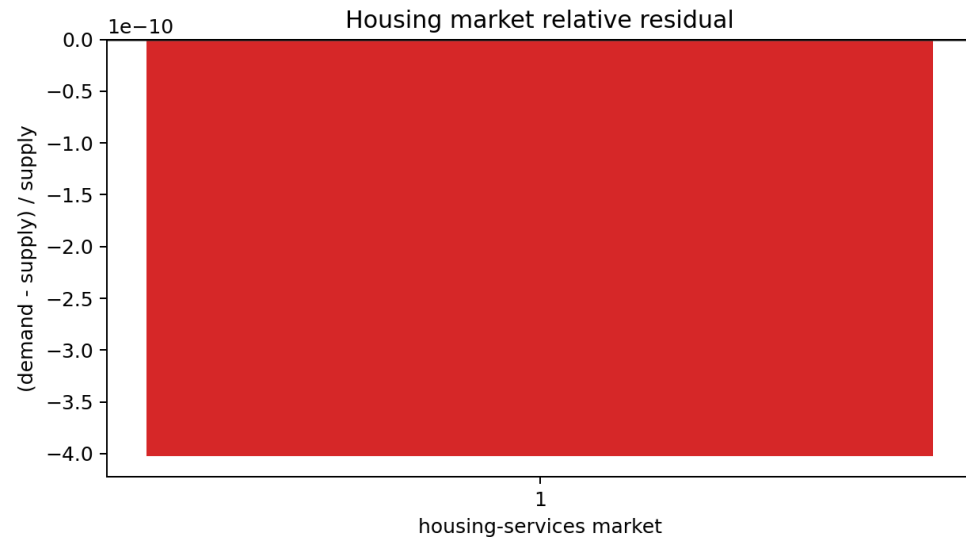
market clearing by market



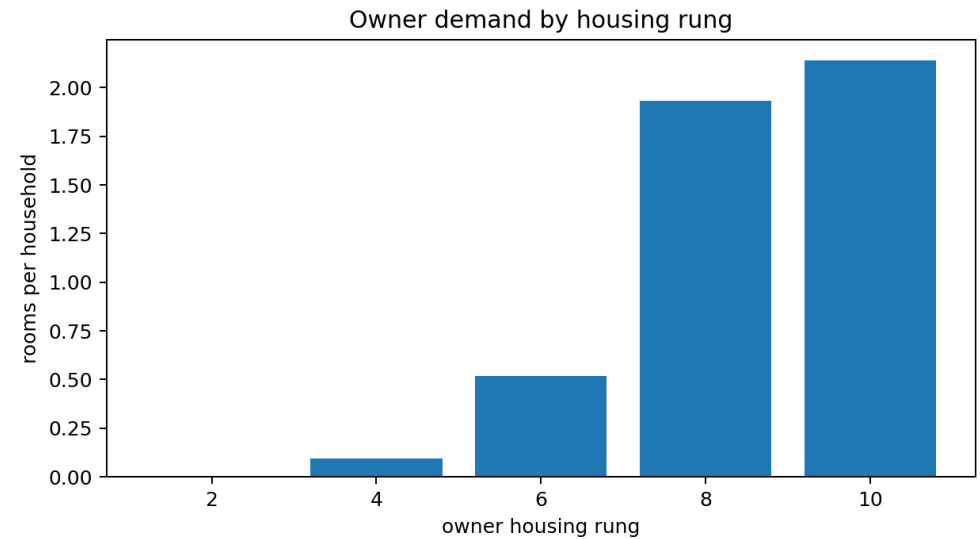
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market clearing residuals



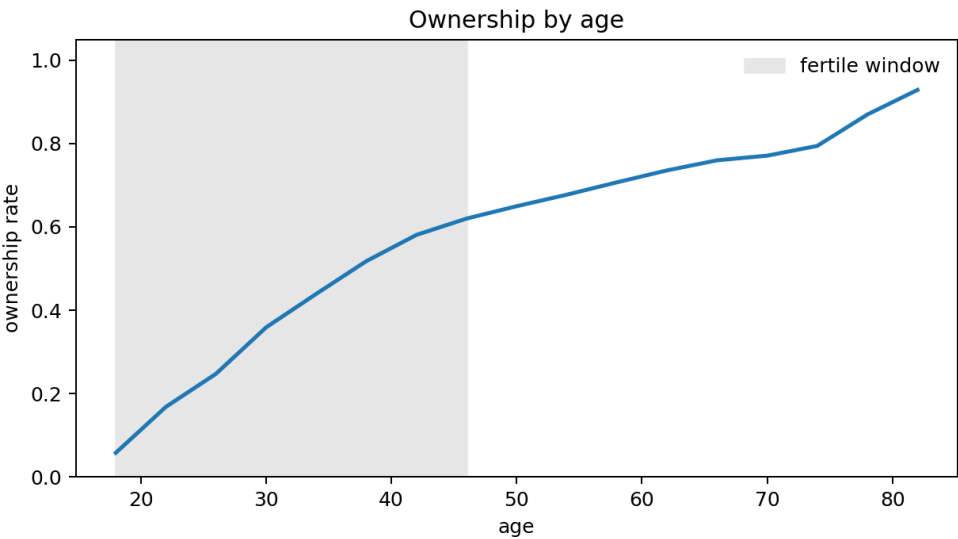
owner rungs



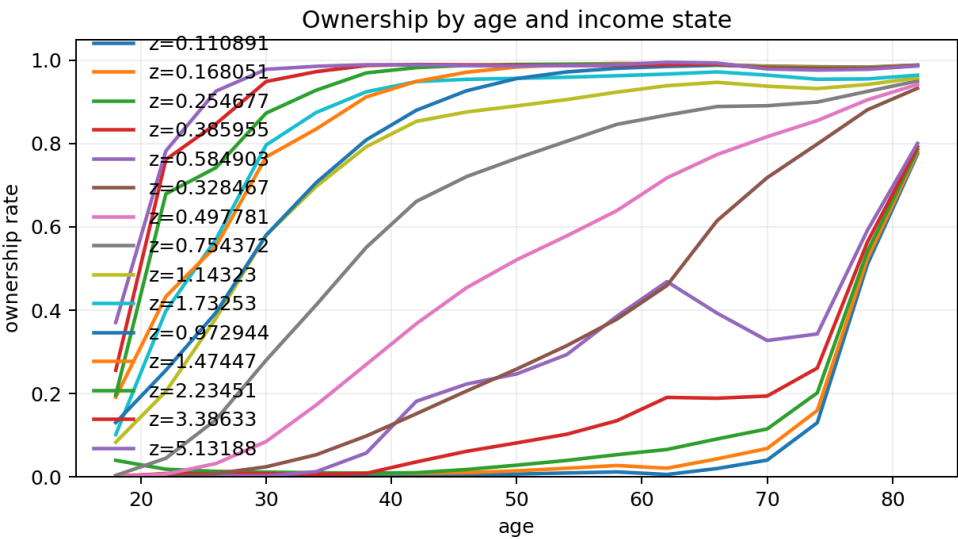
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ownership by age



ownership by age income state

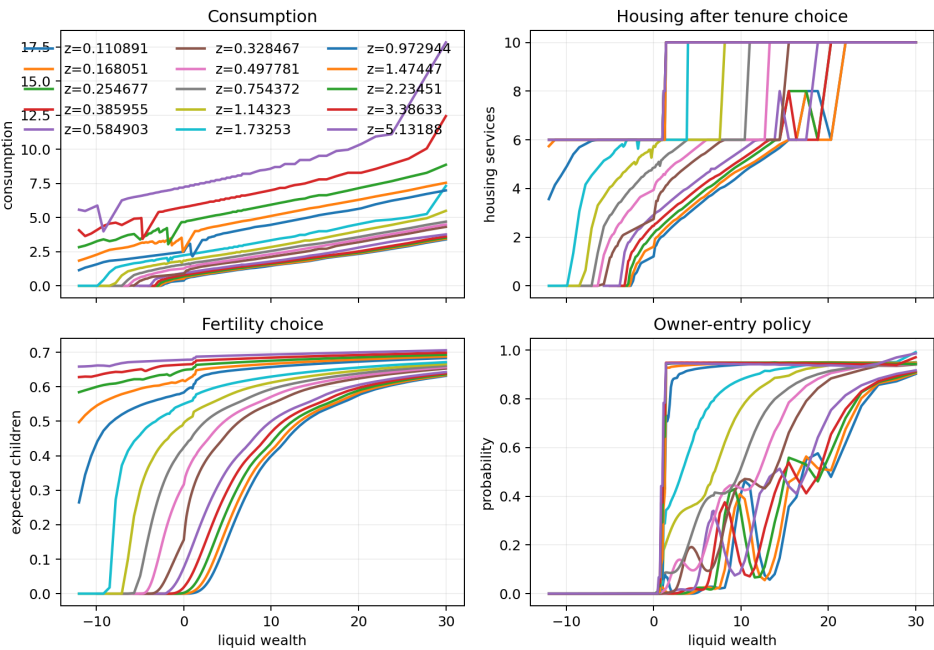


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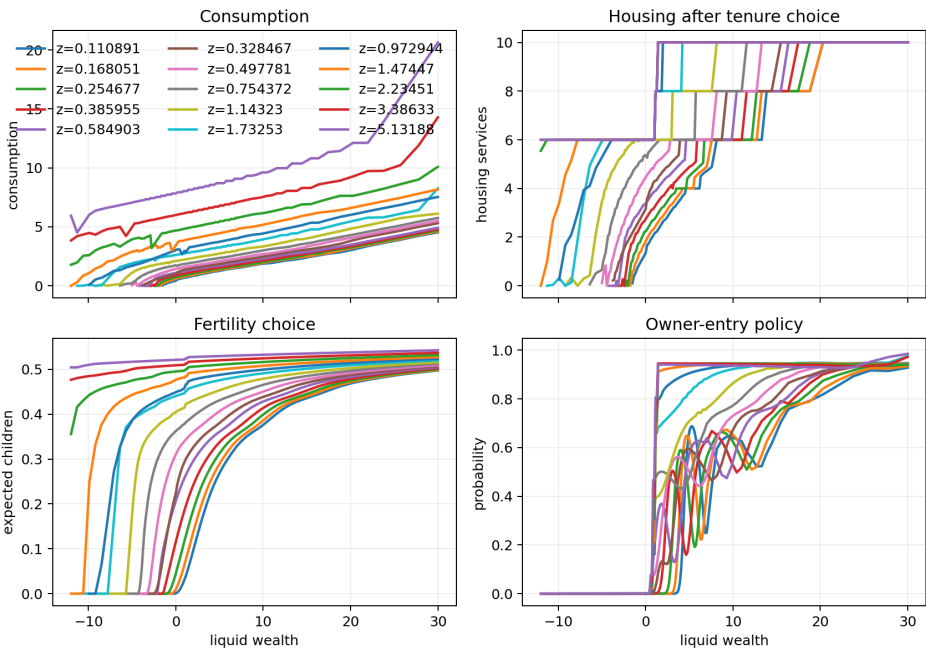
policy childless renter age30

Childless renter policies at age 30



policy childless renter age42

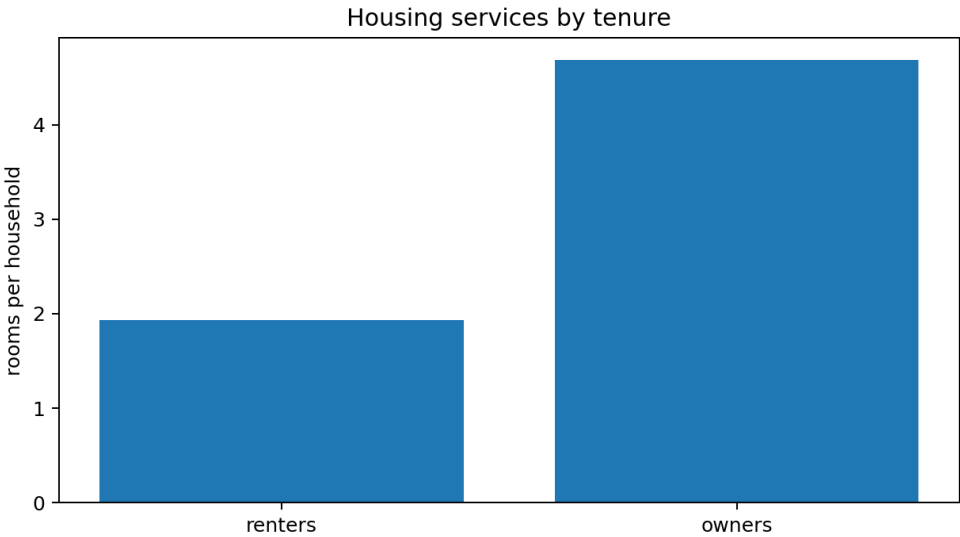
Childless renter policies at age 42



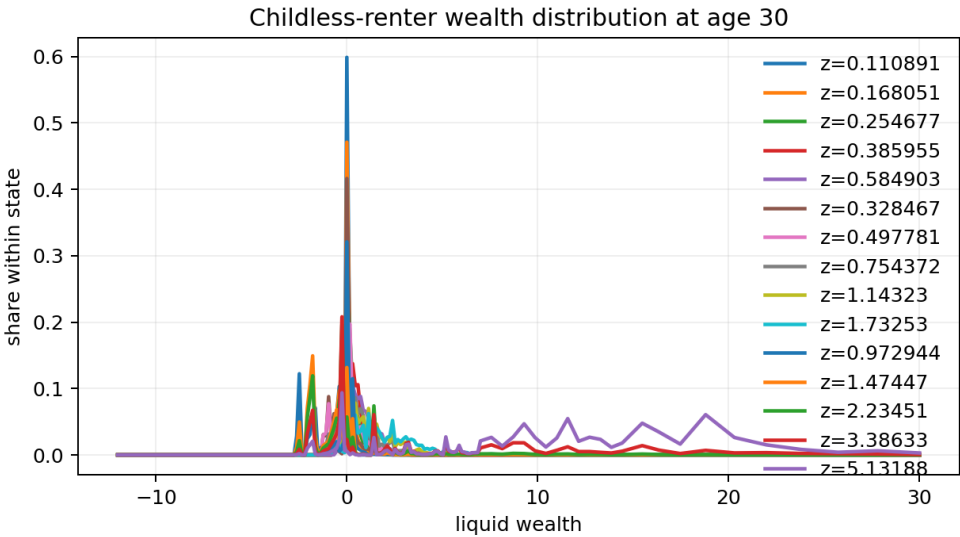
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tenure services



wealth dist childless renter age30



Household diagnostics: 2007 after the first shock

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wealth dist childless renter age42

